Lecture 6

Standard limits and the intermediate value theorem

Uses.

Summary 1: The number $\sqrt{2}$ is irrational.
Summary 2: The Archimedean property supplies integer cutoffs. Bernoulli gives $(1+h)^n \geq 1+nh$ for $h \geq -1$.
Summary 3: Sequence limits obey arithmetic laws. Geometric powers satisfy $q^n \to 0$ for $|q| < 1$.
Summary 4: Bounded monotone sequences converge. The sequences $u_n = (1+1/n)^n$ and $v_n = (1+1/n)^{n+1}$ satisfy $u_n \leq e \leq v_n$ and converge to e, with $2 < e < 3$.
Summary 5: Function limits obey arithmetic, squeeze and qualified composition laws. Continuity preserves sequence limits. In radians, $\sin x/x \to 1$ as $x \to 0$.
Standing fact E: School logarithm, exponential and positive-power laws, monotonicity and continuity are assumed. Angles are in radians.

Objectives. *Standard limits describe elementary functions close to zero. We prove the logarithmic and exponential limits from the construction of e, before using any derivative. We then prove that a continuous real function on an interval takes every value between two of its values. This gives a method for locating an unknown root with a certified error.*

By the end of this lecture you will be able to:

- Derive the logarithmic, exponential and real-power standard limits.
- Use asymptotic equivalence in products and quotients with valid domains.
- Evaluate positive-base power limits through logarithms.
- Prove existence of a root by continuity and bound its location by bisection.

6.1 The logarithmic standard limit

For functions on a real interval, *strictly increasing* means $f(u) < f(v)$ whenever $u < v$ are domain inputs. *Strictly decreasing* reverses this inequality. *Nondecreasing* permits equality, $f(u) \leq f(v)$ for $u < v$. *Nonincreasing* reverses this last inequality.

The natural logarithm $\ln : (0,\infty) \to \mathbb{R}$ is the school inverse of $x \mapsto e^x$. Its order and power laws are part of standing fact E. The bounds on e proved in Lecture 4 give more information than its numerical value. Applying the increasing logarithm to those bounds gives

$$n \ln(1+1/n) \leq 1 \leq (n+1)\ln(1+1/n).$$

Thus the logarithm at reciprocal inputs satisfies

$$\frac{1}{n+1} \le \ln(1+1/n) \le \frac{1}{n} \qquad (n \in \mathbb{N}). \tag{6.1}$$

We use these bounds to control every sufficiently small input, not only reciprocals of integers.

Theorem 6.1 (Logarithmic standard limit). *On the domain $x > -1$, with $x \neq 0$, we have*

$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1.$$

Proof. First let $0 < t < 1/2$. Choose the integer $n = \lfloor 1/t \rfloor$, where $\lfloor r \rfloor$ is the greatest integer at most r. Then $1/(n+1) < t \le 1/n$. Monotonicity of the logarithm and Eq. (6.1), applied at $n+1$ and at n, give

$$\frac{1}{n+2} \le \ln(1+1/(n+1)) \le \ln(1+t) \le \ln(1+1/n) \le \frac{1}{n}.$$

The inequalities $n \le 1/t < n+1$ now imply

$$\frac{n}{n+2} \le \frac{\ln(1+t)}{t} \le \frac{n+1}{n}.$$

For an explicit error test, choose an integer $N > \max\{2, 2/\varepsilon\}$. If $0 < t < 1/(N+1)$, then $n > N$. The lower bound differs from 1 by $2/(n+2) < \varepsilon$. The upper bound differs from 1 by $1/n < \varepsilon$. This proves the right-hand limit.

For $-1 < t < 0$, put $s = -t/(1+t) > 0$. As $t \to 0-$, the quotient law gives $s \to 0+$, and $t = -s/(1+s)$. The logarithm law gives

$$\frac{\ln(1+t)}{t} = (1+s)\frac{\ln(1+s)}{s} \longrightarrow 1.$$

The inner argument s is positive, so the proved right-hand limit applies. Both one-sided limits equal 1. Given a tolerance, the smaller of the two one-sided radii controls every nonzero input near zero. Hence the two-sided limit has the stated value. ∎

For example, on its natural domain near zero,

$$\frac{\ln(1+3x)}{x} = 3\frac{\ln(1+3x)}{3x} \longrightarrow 3.$$

The argument $3x$ tends to zero and is nonzero when $x \neq 0$. The domain condition $1+3x > 0$ holds throughout a neighbourhood of zero.

6.2 Exponential, power and trigonometric limits

Theorem 6.2 (Exponential and real-power standard limits). *For every fixed $\alpha \in \mathbb{R}$,*

$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x\to 0} \frac{(1+x)^\alpha - 1}{x} = \alpha.$$

The power expression uses $x > -1$ and $x \neq 0$.

Proof. Set $u = e^x - 1$. School continuity gives $u \to 0$ as $x \to 0$. Strict increase of the exponential gives $u \neq 0$ when $x \neq 0$. Since $x = \ln(1+u)$, the logarithmic standard limit yields

$$\frac{e^x - 1}{x} = \left(\frac{\ln(1+u)}{u}\right)^{-1} \longrightarrow 1.$$

The denominator tends to 1 and is nonzero near zero, so taking its reciprocal is justified.

For $\alpha = 0$, the second quotient is identically zero. For $\alpha \neq 0$, use the positive-power law $(1+x)^\alpha = e^{\alpha \ln(1+x)}$. The inner exponent tends to zero and is nonzero for $x \neq 0$. Factoring gives

$$\frac{(1+x)^\alpha - 1}{x} = \frac{e^{\alpha \ln(1+x)} - 1}{\alpha \ln(1+x)} \, \alpha \, \frac{\ln(1+x)}{x} \longrightarrow 1 \cdot \alpha \cdot 1.$$

Each factor has a proved finite limit. This proves the stated value. ∎

For example, the positive square root is the power with $\alpha = 1/2$. Consequently $(\sqrt{1+x} - 1)/x \to 1/2$. This also follows by rationalisation, which gives the same value without a power law. For an exponential with base $a > 0$, the identity $a^x = e^{x \ln a}$ gives $(a^x - 1)/x \to \ln a$. If $a = 1$, both sides are zero. If $a \neq 1$, the exponent $x \ln a$ is nonzero on the deleted domain, so the exponential standard limit proves the assertion by the same factorisation.

Two further limits follow from Lecture 5's sine limit and school identities:

$$\frac{\tan x}{x} \longrightarrow 1, \qquad \frac{1 - \cos x}{x^2} \longrightarrow \frac{1}{2} \qquad (x \to 0). \tag{6.2}$$

For the first, write $\tan x / x = (\sin x / x)/\cos x$. The denominator tends to 1 and is nonzero near zero. For the second, the identity $1 - \cos x = 2 \sin^2(x/2)$ gives

$$\frac{1 - \cos x}{x^2} = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2 \longrightarrow \frac{1}{2}.$$

These arguments use neither a differentiation rule nor a series expansion.

6.3 Asymptotic equivalence

Definition 6.3 (Asymptotic equivalence)**.** For real functions f and g on a common domain with cluster point c, write $f(x) \sim g(x)$ as $x \to c$ if g is nonzero throughout some deleted neighbourhood of c and $f(x)/g(x) \to 1$ there.

In particular, the standard limits say

$$\ln(1+x) \sim x, \qquad e^x - 1 \sim x, \qquad \sin x \sim x, \qquad 1 - \cos x \sim x^2/2 \qquad (x \to 0).$$

Each comparison function is nonzero on the deleted domain. If $f \sim g$, the ratio is eventually positive and nonzero because it tends to 1. Thus f is also eventually nonzero. The statement $f \sim g$ alone does not require that either function tend to zero. Constants also qualify.

Equivalents may be multiplied and divided. More precisely, suppose $f \sim g$ and $h \sim k$ on the same approach domain. The quotient identities

$$\frac{fh}{gk} = \frac{f}{g}\frac{h}{k} \longrightarrow 1, \qquad \frac{f/h}{g/k} = \frac{f}{g}\frac{k}{h} \longrightarrow 1$$

prove $fh \sim gk$ and $f/h \sim g/k$. All denominators are nonzero near the approach point by the preceding observation. If g/k has a finite limit, the identity $f/h = ((f/h)/(g/k))(g/k)$

shows that f/h has the same limit. This is the precise justification for replacing factors in a quotient.

For example, consider

$$H(x) = \frac{(e^{2x} - 1)\sin(3x)}{x\ln(1+4x)}.$$

For $0 < |x| < 1/8$, the logarithm has positive argument and is nonzero. Factoring into standard quotients gives

$$H(x) = \frac{3}{2}\,\frac{e^{2x}-1}{2x}\,\frac{\sin(3x)}{3x}\left(\frac{\ln(1+4x)}{4x}\right)^{-1} \longrightarrow \frac{3}{2}.$$

Each inner input tends to zero and avoids zero. The four constants and powers of x have been retained in the factorisation.

Subtraction does not have the same replacement rule. The functions $f(x) = x + x^2$ and $g(x) = x$ satisfy $f \sim g$ at zero, but $(f-g)/x^2 = 1$ for $x \neq 0$. Replacing f by g in this difference would give zero. Cancellation can make a smaller term determine the answer.

6.4 Positive-base power limits

For real functions a and b with $a(x) > 0$, the school power law gives

$$a(x)^{b(x)} = \exp\big(b(x)\ln a(x)\big).$$

If the exponent product tends to a finite number ℓ, continuity of exp gives the limit e^{ℓ}. This method applies even when the separate base and exponent limits would form an indeterminate pattern. For example, as $x \to 0$ with $x > -1$ and $x \neq 0$,

$$(1+x)^{1/x} = \exp\left(\frac{\ln(1+x)}{x}\right) \longrightarrow e.$$

The positive-base hypothesis is essential to this real logarithm argument.

An exponent tending to $+\infty$ gives an exponential tending to $+\infty$. Indeed, given $M > 0$, choose an integer $N > M$. Since $e > 2$, Bernoulli gives $e^N \geq 1 + N(e-1) > N > M$. For every $t > N$, monotonicity gives $e^t > M$. For $t \to -\infty$, writing $e^t = 1/e^{-t}$ gives a limit of zero: for a requested $\varepsilon > 0$, make $e^{-t} > 1/\varepsilon$. These conclusions also apply to a variable exponent by its limit definition.

The following sequences show why the power patterns alone do not specify a limit. Their bases are positive for all $n \in \mathbb{N}$.

Pattern	One outcome	Another outcome
0^0	$(e^{-n})^{1/n} = e^{-1}$	$(e^{-n})^{1/n^2} = e^{-1/n} \to 1$
∞^0	$(e^{n})^{1/n} = e$	$(e^{n})^{1/n^2} = e^{1/n} \to 1$
1^∞	$(1+1/n)^n \to e$	$1^n = 1$

In each row the component limits agree while the resulting limits differ. All three pattern labels describe these component limits. They are not substitutions into the expression being evaluated.

6.5 Intermediate values and bisection

Theorem 6.4 (Intermediate value theorem). *Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If $y \in \mathbb{R}$ lies between $f(a)$ and $f(b)$, including either endpoint value, then there is $c \in [a, b]$ with $f(c) = y$.*

Proof. If y equals an endpoint value, that endpoint supplies c. If both endpoint values agree, this also covers the only possible y. Replacing f and y by their negatives handles decreasing endpoint order. It therefore remains to consider $f(a) < y < f(b)$.

Start with $a_0 = a$ and $b_0 = b$. At stage n, take the midpoint $m_n = (a_n + b_n)/2$. If $f(m_n) = y$, the conclusion is proved. If $f(m_n) < y$, retain $[m_n, b_n]$. Otherwise retain $[a_n, m_n]$. If no stage gives equality, this constructs nested closed intervals with

$$f(a_n) < y < f(b_n), \qquad b_n - a_n = \frac{b - a}{2^n}.$$

The left endpoints are nondecreasing and bounded above by b. Monotone convergence gives $a_n \to c \in [a, b]$, using order to retain the original endpoint bounds. The lengths tend to zero, so $b_n \to c$. Continuity gives $f(a_n) \to f(c)$ and $f(b_n) \to f(c)$. Passing both inequalities to the limit gives $f(c) \le y \le f(c)$. Thus $f(c) = y$, as required. ∎

The domain condition matters. The function $r(x) = x^2 - 2$ restricted to $\mathbb{Q} \cap [1, 2]$ is continuous relative to that domain. Its endpoint values are -1 and 2. It has no zero in the domain, since a rational square root of 2 would contradict Lecture 1. This domain is not a real interval.

The theorem also shows that a continuous image of a real interval is an interval. Here an interval means a subset containing every real number between any two of its members. Given two image values, choose domain points producing them and restrict f to the closed interval between those points. The theorem supplies every intermediate image value. The original interval may be open or unbounded. The chosen closed subinterval still lies in it.

The bisection construction also gives an error bound for an unknown root. After k halvings, a retained interval has length $(b - a)/2^k$. If it contains the root c, its midpoint z_k satisfies

$$|z_k - c| \le \frac{b - a}{2^{k+1}}. \tag{6.3}$$

Both distances from the midpoint to its endpoints equal the right-hand side. Every point of the retained interval is within this half-length of its midpoint. Applying this to the root gives the displayed error bound.

Figure 6.1 shows the retained intervals in the following example.

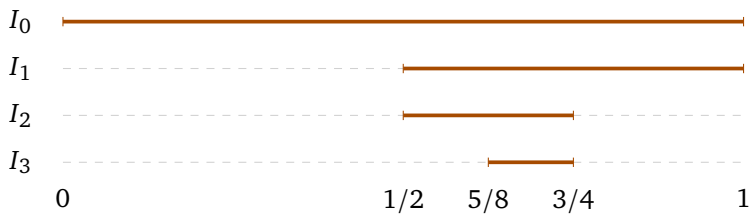


Figure 6.1: Closed intervals retained for the cubic root below. Each halving keeps opposite signs at its endpoints. The common root stays in every interval.

Example 6.5 (A unique root with an error bound). Find a certified interval for a root of $p(x) = x^3 + x - 1$ on $[0, 1]$. The polynomial is continuous. Since $p(0) = -1$ and $p(1) = 1$,

the intermediate value theorem gives a root in $(0, 1)$. It is unique even on $\mathbb{R}$. For $x < y$,

$$p(y) - p(x) = (y - x)(x^2 + xy + y^2 + 1) > 0,$$

because the second factor equals $(x + y/2)^2 + 3y^2/4 + 1 > 0$. Thus two different inputs cannot have the same value zero.

Bisection gives the following exact computations.

Halving	Tested midpoint	Polynomial value	Retained interval
1	1/2	−3/8	[1/2, 1]
2	3/4	11/64	[1/2, 3/4]
3	5/8	−67/512	[5/8, 3/4]

The final midpoint is $11/16$. The retained width is $1/8$, so its error from the unique root is at most $1/16$. To guarantee error at most $\eta > 0$ from the original unit interval, choose k with $2^{-k-1} \le \eta$. This is an error guarantee from continuity and signs, without a formula for the root or a derivative of the polynomial.

Summary 6.

- Theorems 6.1 and 6.2: the standard logarithmic and exponential quotients tend to 1, and the real-power quotient tends to its exponent α. The power uses a positive base.
- Equation (6.2): $\tan x/x \to 1$ and $(1 - \cos x)/x^2 \to 1/2$ in radians as $x \to 0$.
- Asymptotic equivalence means a quotient tending to 1 with a nonzero comparison function near the approach point. It permits multiplication and division, but does not justify cancellation in a difference.
- For positive bases, $a^b = \exp(b \ln a)$ reduces a power limit to an exponent limit. The patterns 0^0, ∞^0 and 1^∞ are indeterminate.
- Theorem 6.4 and Eq. (6.3): a continuous function on a closed real interval takes every intermediate endpoint value. Bisection retains a root with an explicit midpoint error bound.

Lecture 7 proves attainment of extreme values on a closed interval and introduces derivatives. The standard limits above will supply the first elementary derivative formulas. Sheet 6 practises standard limits, asymptotic comparisons and bisection.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, Chapters 4–5.